# Textual Analysis in Finance

## Week 5: Large Language Models

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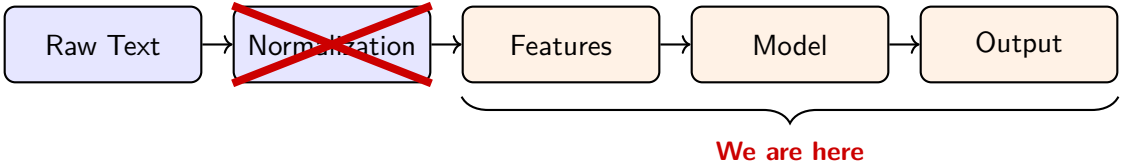
# Where are we?



# Outline

# From LMs to LLMs

# Why do language models get bigger?

- By 2019-2020, empirical evidence suggested that **larger models consistently outperform smaller ones**
  - $\text{BERT}_{\text{LARGE}}$ (340M parameters) outperforms $\text{BERT}_{\text{BASE}}$ (110M parameters) across a wide range of tasks (e.g., sentiment analysis, question answering, natural language inference) [Devlin et al., 2019]
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- **Performance improvements were systematic**: test loss decreases with increases in model size, data size, and compute [Kaplan et al., 2020]
- This naturally led to aggressive scaling along three dimensions:
  - **Model size:** Millions $\rightarrow$ billions (and beyond) of parameters
  - **Training data:** Gigabytes $\rightarrow$ terabytes $\rightarrow$ internet-scale corpora
  - **Compute:** Massive distributed GPU/TPU clusters enabling large-scale optimization

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→ Scaling is no longer purely a modeling question, it becomes an **economic one**

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→ The view of LMs shifts from **representation learning** to a **general cognitive interface** (or text generation)

# Training an LLM

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- RLHF provides **human feedback to guide the model** toward helpful, safe, and aligned responses

# RLHF - Analogy to raising kids

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  - Raising kids: The kid learns from everything they observe (books, conversations, TV) without guidance
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- **Outcome = Aligned behavior**
  - Raising kids: With consistent feedback, the kid learns to behave thoughtfully
  - RLHF analogy: LLMs trained with RLHF generate responses that are truthful, helpful, and aligned with user expectations

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  2. Fine-tune the LLM on the human-based prompt-feedback dataset (called **supervised fine-tuning**)
- Collect comparison data and train a reward model
  1. Collect more prompts and the corresponding LLM's responses (several responses per prompt)
  2. Ask humans again to rank the preference of those responses
  3. Use those prompts, responses, and human ranking to train another small LM, called the **reward model**
  4. The reward model outputs a score per response (no text generation), in which more preferred responses receive higher scores

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- Reinforcement learning happens here:
  1. Feed new prompts to the LLM
  2. The LLM generates (usually) one response per prompt
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- PPO pushes the LLM toward higher-reward responses **gradually** (i.e., it does not move "far" from the previous version)
  - Like teaching a robot to cook for someone who wants to gain weight
  - If we force the robot to get it perfect in one step, it might just hand us a bottle of cooking oil
  - PPO forces the robot to learn gradually: make a first dish, tweak the salt or sugar, get feedback, repeat step by step

# Text generation

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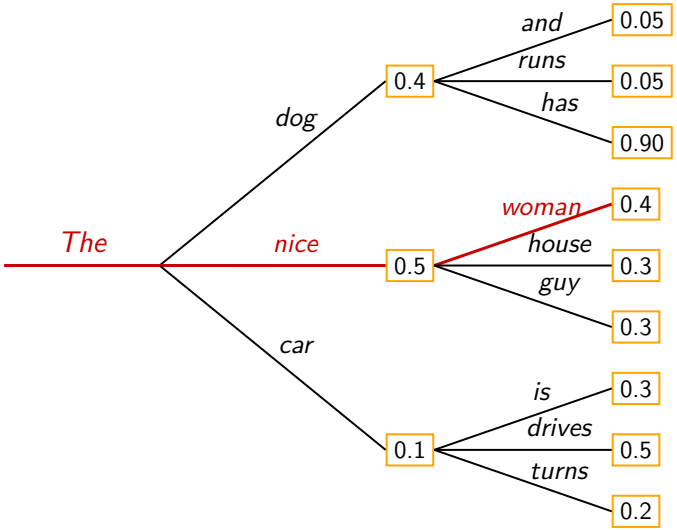
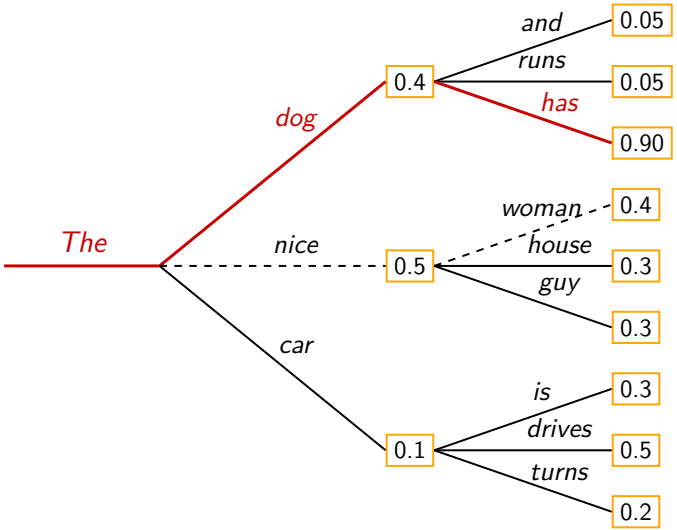
- A great blog about text generation can be found here

# Greedy search



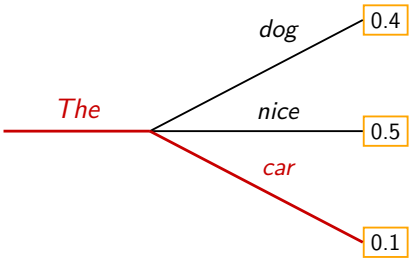

- Greedy search, a.k.a Just go for the one with the **highest probability**
- Start with "The" $\rightarrow$ "nice" $\rightarrow$ "woman"
- Main drawback: *it misses high probability words hidden behind a low probability word*
- The probability of the whole generated text is $0.5 \times 0.4$
- "The" $\rightarrow$ "dog" $\rightarrow$ "has" is a better choice!

# Beam search



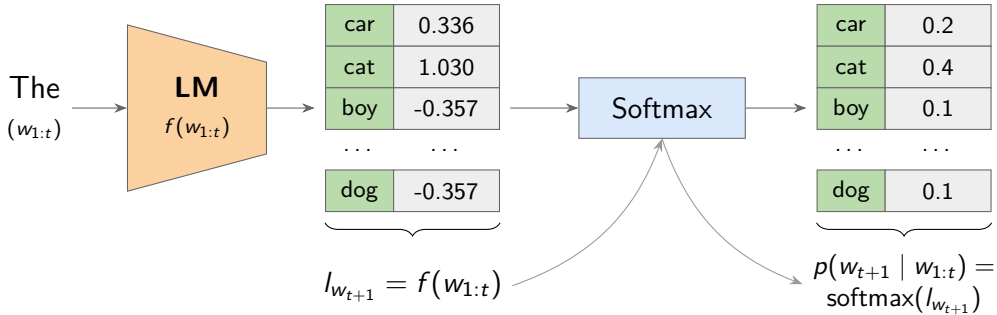

- The probability of the full sequence can be higher than in greedy search: $0.4 \times 0.9 > 0.5 \times 0.4$
- In this case, beam width $= 2$ (number of candidate sequences kept at each step)
- Controlled by `num_beams` in the `.generate()` method
- Drawbacks: Generated texts often ...
  - contain repetitions
  - lack of diversity [Holtzman et al., 2019]

# Sampling



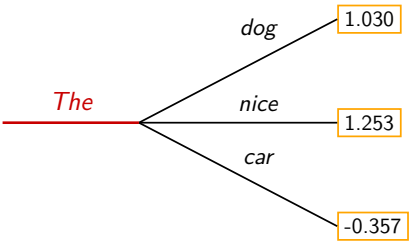

- Purpose: introduce stochasticity in text generation
- Instead of choosing $w_{t+1} = \arg\max_w \hat{p}(w \mid w_{1:t})$, sample the next token $w_{t+1} \sim \hat{p}(w \mid w_{1:t})$
- Low-probability tokens can occasionally be selected (e.g., "car") $\rightarrow$ more diverse and creative text
- The randomness is controlled by `set_seed` and `do_sample`
- Drawback: Generated texts can be often gibberish, meaning, they don't make sense

# Compute $p(w_{t+1}|w_{1:t})$




- Given the context $w_{1:t}$, an LM will compute "logits" of the next token, $l_{w_{t+1}}$
- Basically, each token in the vocabulary has a logit value
- Softmax is a function that transforms logits, a list of numbers, into probabilities, i.e., also a list of numbers, but bounded between 0 and 1, and sum up to 1

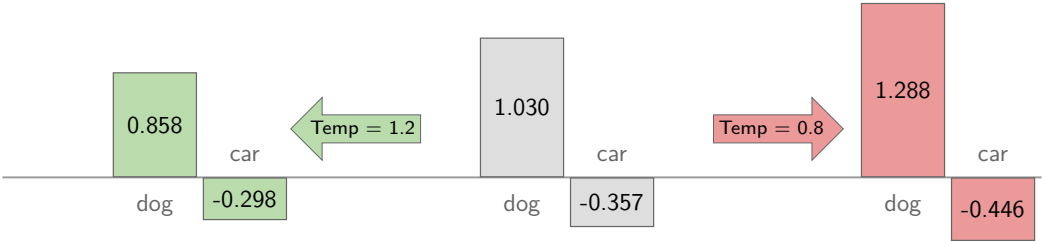
# Sampling - Temperature




How does Softmax work in this case?

- Assume the vocabulary contains only 3 tokens: dog, nice, and car
- Softmax first computes $e^{1.030}$, $e^{1.253}$, and $e^{-0.357}$ for each word
- Then, $p(w_{t+1} = \text{dog}|w_{1:t}) = \frac{e^{1.030}}{e^{1.030}+e^{1.253}+e^{-0.357}} \approx 0.4$
- Or, $p(w_{t+1} = \text{dog}|w_{1:t}) = \frac{e^{\frac{1.030}{\mathbf{1}}}}{e^{\frac{1.030}{\mathbf{1}}}+e^{\frac{1.253}{\mathbf{1}}}+e^{\frac{-0.357}{\mathbf{1}}}}$
- The $\mathbf{1}$'s are called **Temperature**, and used to control the distribution of the next token
- The name "Temperature" comes from Boltzmann distribution in thermodynamics

# The effect of "Temperature"

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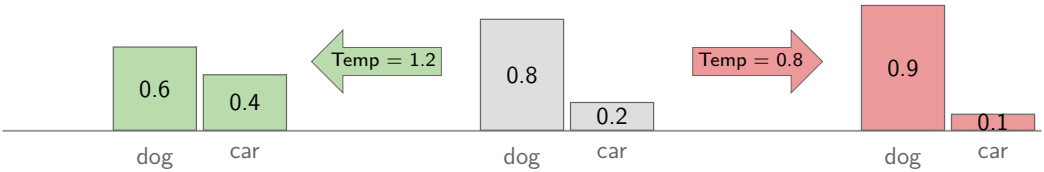


- Temperature scales the logits and **changes the spread of probabilities**
- **High temperature** → smaller logit differences → flatter distribution
- **Low temperature** → larger logit differences → shaper distribution

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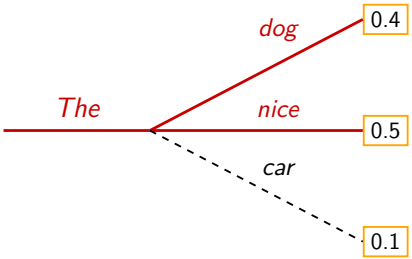
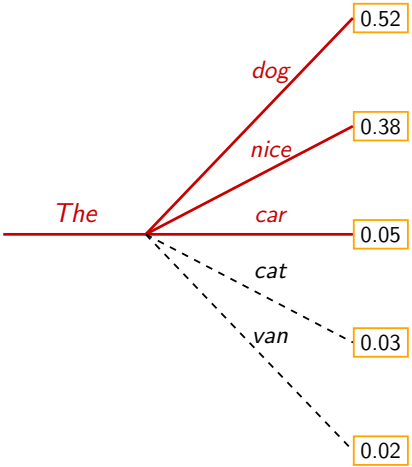
- Similarly, **low temperature** $\rightarrow$ shaper $p(w_{t+1}|w_{1:t}) \rightarrow$ generated texts are **more conservative** (more deterministic)
- More on empirical insights:
  - Temperature is not a "one-size-fits-all" hyperparameter [Li et al., 2025], but DeepSeek recommend some default values depending on the tasks here
  - $T = 0 \rightarrow$ greedy decoding
  - "*...the influence of temperature on creativity is far more nuanced and weak than suggested by the 'creativity parameter' claim...*" [Peeperkorn et al., 2024]
  - Creativity is controlled by many other hyperparameters, not just temperature

# Top-$K$ sampling




- Introduced by Fan et al. [2018], simple yet efficient
- Restrict sampling only from the top $K$ tokens with the highest probabilities
- Probabilities are re-distributed: $(0.5, 0.4) \rightarrow (0.556, 0.444)$
- $K \uparrow \rightarrow$ more diversity; $K \downarrow \rightarrow$ more conservative outputs
- Typically applied after temperature scaling

# Top-$p$ (nucleus) sampling




- Top-$p$ (nucleus) sampling: select the **smallest set of tokens** whose **cumulative probability** $\geq p$
- Example: $p = 0.94 \rightarrow$ keep "dog", "nice", "car" since $0.52 + 0.38 + 0.05 = 0.95$
- Can be combined with Top-$K$:
  - Apply Top-$K$ first, then Top-$p$ source code
  - Final set satisfies **both constraints**

# Why LLMs hallucinate?

- Hallucination: LLMs produce "plausible yet incorrect statements instead of admitting uncertainty" [Kalai et al., 2025]
- LLMs hallucinate, even with the latest model like GPT-5
- But why? Hallucination roots from all steps:
  - Pre-training
  - Calibration by RLHF

# Hallucination from pre-training

- Even with error-free training data, **the statistical objective** minimized during pretraining would lead to an LLM that generates errors
- For example:
  - The sentence "Lionel Messi is a Spanish football player" is correct linguistically, but it's incorrect as a fact
  - The current statistical objectives only care about linguistic correctness, not facts
  - An LLM trained on this data, by the current statistical objective, will hallucinate; i.e., it thinks it is producing a correct response
- Other reasons:
  - Poor models
  - Distribution shift: Users ask questions that have been never seen before in the training data
  - Garbage in, Garbage out: Training data usually contains factual errors as well

# Hallucination from the post-training calibration

- Recall: in RLHF, the reward model (teacher) score the response from the LLM (student) given a prompt
- This motivates guessing when the LLM faces uncertainty
- Analogy:
  - Students are having a multiple choice question in an exam
  - The question has 3 choices: one true, one false, and one "I don't know"
  - We rarely choose "I don't know" as it sounds like an "always wrong" answer $\rightarrow$ it maximally penalize the overall scores
  - It's because doing so **maximizes the expected utility**
- LLMs act the same because we evaluate them as if they are taking exams
- Or, the way we train and evaluate LLMs encourages them guessing, making them hallucinate

# Coming next

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- Hands-on practice
  - Text generation
  - Prompt engineering
  - Notebook/Solution
- Next lecture
  - Post-training adaptation
  - Course feedback and discussion
  - Guest lecture

# References I


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